

WASP-12b

R-0280

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_241001 · created 2026-07-17T15:13:16+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460584.9075 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1282 +/- 0.0095
Transit depth (Rp/Rs)^2	1.64 +/- 0.24 [%]
Orbital Inclination (inc)	82.57 +/- 1.89
Airmass coefficient 1 (a1)	1.218 +/- 0.013
Airmass coefficient 2 (a2)	-0.1833 +/- 0.0037
Scatter in the residuals of the lightcurve fit is	1.03 %
Best Comparison Star	None
Optimal Aperture	4.67
Optimal Annulus	10.14
Transit Duration (day)	0.1237 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1282 ± 0.0095 vs 0.1178 (deviation 1.07σ)
Duration fit vs published	0.1237 d vs 0.125 d (deviation 0.3σ)
Mid-time O–C	-1.7 min
Depth SNR	13.2
Residual scatter	1.03 %

Light curve

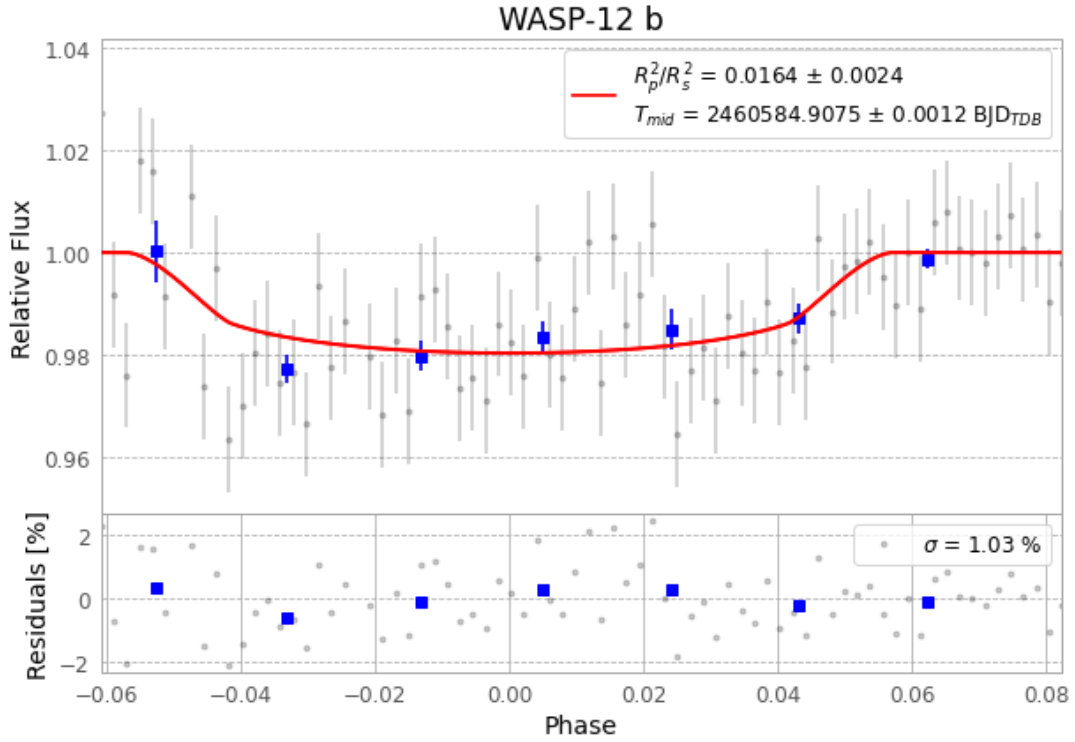


Figure 1 — FinalLightCurve_WASP-12 b_2024-10-01.png (SHA-256 b6cdb26163021d49...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [340, 181], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d89ec85ff8b38ef8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit faa81f0

Data provenance

76 FITS frames (50 MB), WASP-12241001080915.FITS → WASP-12241001115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-241001.log (a6f286cac8e6...), FinalLightCurve_WASP-12 b_2024-10-01.png (b6cdb2616302...), FinalLightCurve_WASP-12 b_2024-10-01.csv (b8f48a99f25e...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 15:13Z.